

Multi-Source Candidate Data Transformer — Design

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PROBLEM FRAMING

Eightfold receives candidate data from many sources at once, in conflicting shapes and formats. Downstream needs **one canonical profile per candidate**: fixed fields, normalized formats, deduplicated, with provenance and confidence. Guiding principle: **wrong-but-confident is worse than honestly-empty** — unknowns become `null`, never invented.

PIPELINE

detect → extract → normalize → merge → confidence → project → validate

- detect** — identifies each input's source type (explicit override or inferred).
- extract** — a per-source adapter (common interface) maps the source's shape to our fields. No normalization or merging here.
- normalize** — pure functions: phones→E.164, dates→YYYY-MM, country→ISO-3166 alpha-2, skills→canonical, emails lowercased/validated. Unparseable → `null` (raw kept in provenance).
- merge** — clusters records by blocking keys (email, phone, name+location) via union-find, then resolves each field to a winner.
- confidence** — scores each value and the overall profile.
- project** — applies the runtime config to shape output from the canonical record.
- validate** — checks output against the requested schema; degrades gracefully.

CANONICAL SCHEMA & NORMALIZED FORMATS

Fields: `candidate_id`, `full_name`, `emails[]`, `phones[]`, `location{city,region,country}`, `links{linkedin,github,portfolio,other[]}`, `headline`, `years_experience`, `skills[{name,confidence,sources[]}]`, `experience[{company,title,start,end,summary}]`, `education[{institution,degree,field,end_year}]`, `provenance[{field,source,method,value}]`, `overall_confidence`.

Formats: phones E.164, dates YYYY-MM, country ISO-3166 alpha-2, skills canonical (dictionary), emails lowercased+validated. `candidate_id` = stable hash of the strongest key (email → phone → name|company).

MERGE / CONFLICT RESOLUTION

Source trust order: `recruiter_csv` > `ats_json` > `github` > `resume` > `recruiter_notes`. Records cluster by shared normalized email/phone (and weakly, name+country — **never name alone**, to avoid over-merging). Each field's winner: **source trust** → **cross-source agreement** → **completeness** → **recency**, ties broken deterministically (trust, source name, input order). Multi-valued fields are unioned, de-duplicated, ordered by trust. Every contributing value is recorded in provenance.

CONFIDENCE

`field_conf` = `base_trust` × `method_factor` × `agreement_boost` × `normalization_factor`, clamped to [0,1]. LLM/notes values get lower method and trust factors so they cannot silently overpower structured data.
`overall_confidence` = importance-weighted mean over **populated** fields only (honestly-empty isn't penalized as wrong).

RUNTIME CUSTOM-OUTPUT CONFIG

A clean projection layer, separate from the canonical record, reads a config that can: select a subset of fields, rename/remap via `from` paths (`emails[0]`, `skills[].name`), set per-field normalization, toggle provenance/confidence, and choose `on_missing` = `null` | `omit` | `error`. The projected output is validated against the schema implied by the config. **One engine serves many downstream schemas with no code changes.**

EDGE CASES & HANDLING

- Garbage/malformed source** (bad JSON, missing file) → skipped with a warning; run continues on remaining sources.
- Conflicting values** (CSV "Flipkart" vs notes "Amazon") → higher-trust source wins; both retained in provenance; disagreement lowers confidence.
- Same person, no shared email/phone** (only name+initial) → linked only via weak name+location key; never merged on name alone.
- Unparseable value** (phone "call me", date "sometime 2019") → field `null`, raw kept in provenance; never guessed.
- LLM unavailable / no API key** → deterministic regex+dictionary fallback; output stays valid and reproducible.

DELIBERATELY OUT OF SCOPE (TIME PRESSURE)

LinkedIn adapter (designed, not built); full international address parsing; embedding-based skill matching; rich UI. These slot in later without changing merge/projection.

AI USAGE

Groq's free LLM extracts fields from unstructured text with temperature 0 + JSON mode and an explicit "never guess" instruction; a deterministic fallback covers the keyless case. **AI assists extraction only** — formatting, merging, confidence, and validation are deterministic code I can explain end to end.